使用 ONTAP 作为 Kubernetes 的永久性存储

Cloud Manager 3.7

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使用 ONTAP 作□ Kubernetes 的永久性存□

Cloud Manager 可以自□部署 "NetApp Trident" 在 Kubernetes 集群上，以便可以使用 ONTAP 作□容器的永久性存□。□□用于 Cloud Volumes ONTAP 和内部 ONTAP 集群。

在完成□些□□之前，□需要 "□建 Cloud Volumes ONTAP 系□" 或 通□ Cloud Manager 。

如果□使用部署 Kubernetes 集群 "NetApp Kubernetes Service"， Cloud Manager 可以通□□的 NetApp Cloud Central □□自□□□集群。如果是□□情况，□跳□前□个□□，从第 3 □□始。

□□网□□接

1. Cloud Manager 和 Kubernetes 集群之□以及从 Kubernetes 集群到 ONTAP 系□之□必□具有网□□接。
2. 安装 Trident □， Cloud Manager 需要出站 Internet □接才能□□以下端点：

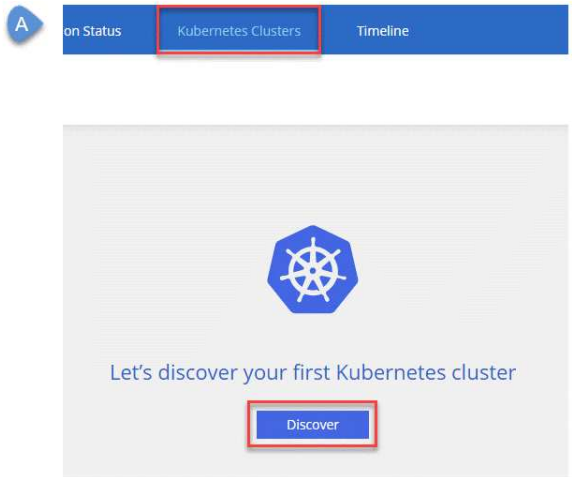
   https://packages.cloud.google.com/yum https://github.com/NetApp/trident/releases/download/

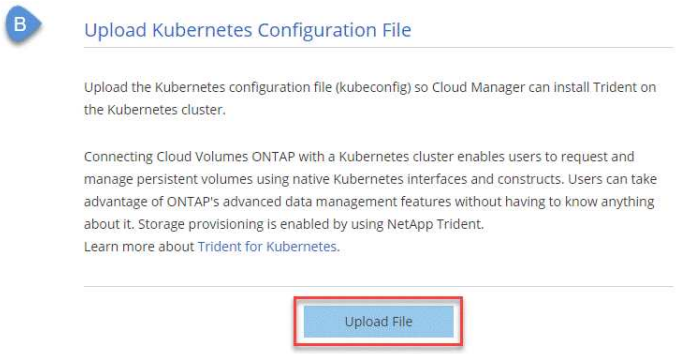
   当□将工作□境□接到 Kubernetes 集群□， Cloud Manager 会在□集群上安装 Trident 。

将 Kubernetes 配置文件上□到 Cloud Manager

□于□个 Kubernetes 集群，□□管理□需要上□一个 YAML 格式的配置文件（ kubeconfig ）。上□文件后，Cloud Manager 会□□与集群的□接，并保存 kubeconfig 文件的加密副本。

□□ * Kubernetes Clusters > Discover > Upload File* ，然后□□ kubeconfig 文件。

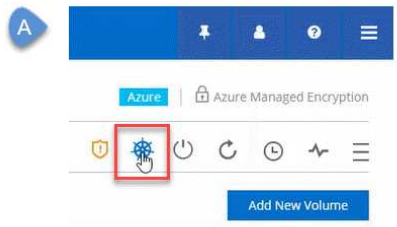


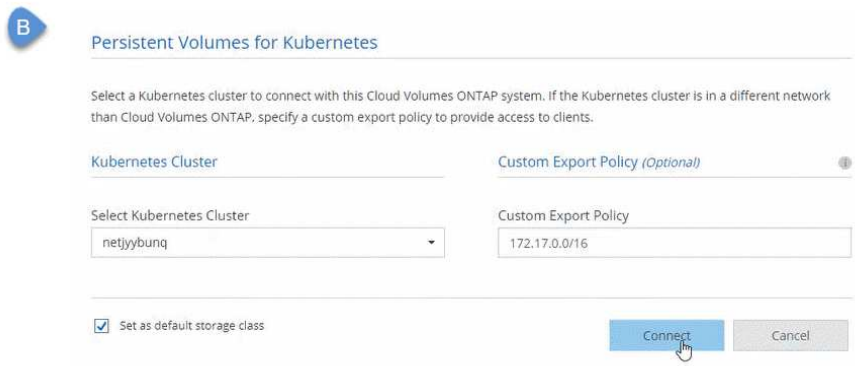


将□的工作□境□接到 Kubernetes 集群

在工作□境中，□□ Kubernetes □□并按照提示□行操作。□可以将不同的集群□接到不同的 ONTAP 系□，将多个集群□接到同一个 ONTAP 系□。

□可以□□将 NetApp 存□□□置□ Kubernetes 集群的默□存□□。默□情况下，当用□□建永久性卷□，Kubernetes 集群可以使用已□接的 ONTAP 系□作□后端存□。





□始配置永久性卷

使用原生 Kubernetes 接口和□造□求和管理永久性卷。Cloud Manager 会□建四个 Kubernetes 存□□，□可以在配置永久性卷□使用□些存□□：

- * NetApp 文件 * ：用于将永久性卷□定到□□点 ONTAP 系□
- * netapp-file-san* ：用于将 iSCSI 永久性卷□定到□□点 ONTAP 系□
- * netapp-file-redundred* ：用于将永久性卷□定到 ONTAP HA □
- * netapp-file-redundred-san-san ：用于将 iSCSI 永久性卷□定到 ONTAP HA □

Cloud Manager 会将 Trident 配置□默□使用以下配置□□：

- 精□卷
- 默□ Snapshot 策略
- 可□□的 Snapshot 目□

"了解有□使用□用于 Kubernetes 的 Trident 配置第一个卷的更多信息"

什□是 trident 卷？

Cloud Manager 会在□接到 Kubernetes 集群的第一个 ONTAP 系□上□建一个卷。卷的名称将附加 "_trident 或 trident 。 " ONTAP 使用此卷□接到 Kubernetes 集群。□不□□除□些卷。

断□或□除 Kubernetes 集群□会□生什□情况？

使用 Cloud Manager ，□可以断□各个 ONTAP 系□与 Kubernetes 集群的□接。断□系□□接后，不能再将□ ONTAP 系□用作容器的永久性存□。不会□除□有永久性卷。

从 Kubernetes 集群断□所有系□的□接后，□□可以从 Cloud Manager 中□除整个 Kubernetes 配置。□除集群□， Cloud Manager 不会卸□ Trident ，也不会□除任何永久性卷。

□□□操作都只能通□ API 来□行。我□□□在未来版本中将□些操作添加到界面中。"□□此□了解有□ API 的□□信息"。

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